PRODUCT REQUIREMENTS DOCUMENT

ePharmatica

The learning & community platform for pharmacy — study smarter, connect professionally, grow your career.



v1.6 · APPROVED FOR BUILD — PHASE 0 / 1

DOMAIN epharmatica.com	DATE 18 July 2026
OWNER Founder / Product Lead	HOSTING Self-hosted VPS · Hetzner EU
STACK Next.js · Supabase · Tailwind · shadcn/ui · Plate	METHOD Structured MCQ elicitation — 77 locked decisions

LEARNING PLATFORM · AI STUDY TOOLS (BYOK) · COMMUNITY & BLOG · JOBS PORTAL · RESUME BUILDER · WHATSAPP ASSISTANT

Confidential — internal planning document

EPHARMATICA · PRD v1.6

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BEFORE YOU READ

ePharmatica at a Glance

2-in-1	77	< $15/mo	5 + 1
Learning platform + professional community	MCQ decisions — 18 PRD · 12 style · 47 module	Total launch infrastructure cost	Delivery phases, each acceptance-gated

One platform, two connected halves. A **learning platform** delivering learning-optimized, structured, visual study material covering the HEC-approved Pharm-D syllabus — block-based written notes paired with short-form (and optionally long-form) videos from the ePharmatica YouTube channel, with accounts for progress, highlights, bookmarks, personal notes, and BYOK-powered AI study tools. And a **professional community** for the pharmacists of Pakistan, open to the world: a hybrid discussion forum, an editorial blog by approved writers, a curated pharmaceutical jobs portal, and an ATS-friendly resume builder. A role-based in-app CMS runs everything; the landing page presents the full vision with unreleased modules marked **Coming Soon**.

STRATEGIC POSTURE

Content-first · cost-disciplined (single self-hosted VPS, free-tier services, BYOK AI — the platform never pays for inference) · globally usable (subject-first curriculum; HEC semester as a subtle tag) · monetization-flexible (no paywall at launch; an entitlement layer is built in so any business model can switch on later without refactoring).

DELIVERY PHASES — each gated by acceptance criteria

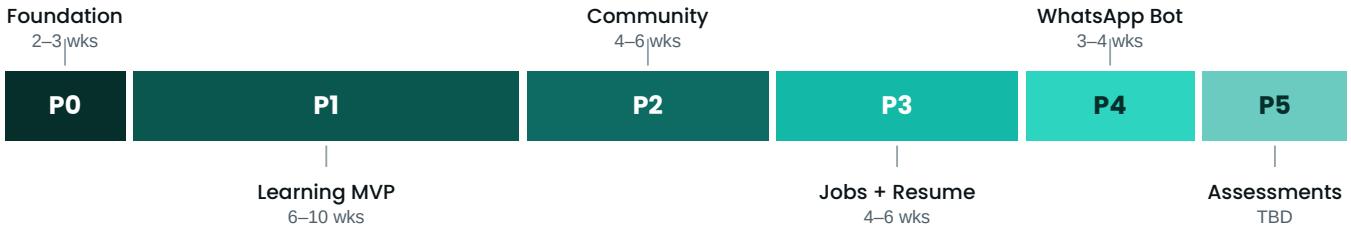


Figure 1 — Phased delivery. P5 scope is confirmed at Phase 3 planning (Assumption A3).

1 Executive Summary

ePharmatica is a web platform with two connected halves:

- **A learning platform** delivering learning-optimized, structured, and visual study material covering the HEC-approved Pharm-D syllabus — written notes authored in a block-based editor, paired with short-form (and optionally long-form) videos embedded from the ePharmatica YouTube channel. Learners create free accounts to track progress, highlight, bookmark, write personal notes, and use AI study tools powered by their own API keys (BYOK).
- **A professional community platform** for pharmacists of Pakistan (open to the world): a hybrid discussion forum, an editorial blog written by approved writers, a curated pharmaceutical jobs portal, and an ATS-friendly resume builder.

Everything is managed through a role-based in-app CMS. The product ships in phases, beginning with the learning system; the public landing page presents the full vision with unreleased modules marked **Coming Soon**.

CHANGELOG

All entries 18 Jul 2026 · **v1.6: module PRD series complete** — eight module PRDs (M1 Learning · M2 Accounts · M3 AI · M4 CMS · M5 Community · M6 Jobs · M7 Resume · M8 WhatsApp), 47 MCQ answers → 51 recorded decisions; **module PRDs are authoritative for module-level detail** (on conflict with §7, they + this changelog govern). Amendments: per-subtopic reader pages (M1-1); subtopic video fields → derived `primary_video_url` (M1-2); jobs launch **curated-via-CMS only**, employer form dormant behind a flag (M6-2); bilingual English/Roman-Urdu WhatsApp bot on a dedicated number (M8-1/2). · **v1.5:** founder style series S1–S12 locked into STYLE.md v1.3 — **0px radius finalized**, 20px gutters / 1200px container, split landing hero over live primitives, always-underlined links, A−/A+ reader sizing, 2px reading bar, instant navigation, Phase-0 tooling (`/dev/system`, Storybook, Style Dictionary → Figma tokens, RTL conventions), **waitlist → email + module + role**. · **v1.4:** STYLE v1.2 — mobile & touch system; specimen copy genericized. · **v1.3: A7 resolved** — Pharmacopoeia palette, monograph-label signature, Archivo / Instrument Sans / Plex Mono (this PDF keeps the interim print face, palette-aligned). · **v1.2:** framework pin corrected to **Next.js 16** (16.2.x); `middleware.ts → proxy.ts`. · **v1.1:** Q1/Q3/Q4/Q5 resolved — epharmatica.com; Hetzner EU; legal drafts; Author-owned semester mapping.

2 Vision, Goals & Non-Goals

2.1 Vision

Become the default study companion for Pharm-D students in Pakistan and a credible professional home for its pharmacists — the place where a student learns pharmacology in first year and, five years later, finds their first industry job.

2.2 Product Goals

- **G1 — Learning quality.** Every published topic must be genuinely easier to learn from than the textbook chapter it maps to (structured, visual, exam-aware, video-paired).
- **G2 — Habit formation.** Students return because their progress, highlights, and notes live here.
- **G3 — Zero-marginal-cost AI.** AI study features at full quality with no inference cost to the platform (BYOK).
- **G4 — Community gravity.** Forum + blog + jobs make ePharmatica valuable after graduation, not just before exams.
- **G5 — Operational frugality.** Total infrastructure cost under ~US$15/month at launch.
- **G6 — Editorial integrity.** All study and blog content passes a review workflow before publication; medical-education accuracy is a hard requirement.

2.3 Non-Goals (explicitly out of scope for v1.x)

- Native mobile apps (the web app is fully responsive; PWA-ready architecture only).
- Live classes, webinars, or cohort courses.

- Hosting video files (all video is embedded from YouTube).
- University-specific LMS features (assignments, attendance, grading).
- Payments/subscriptions at launch (architecture supports them later — §7.11, §16).
- Direct in-platform job applications with employer dashboards (v1 links out to apply; employer self-serve is limited to a submission form).
- Automated pharmacist license verification against regulator databases (v1 verification is manual review).

3 Target Users & Personas

P1 · PRIMARY

Ayesha — 2nd-year Pharm-D student, Lahore

Studies from slides + WhatsApp'd PDFs; loses her notes every semester. Needs clear explanations mapped to her syllabus, videos for hard topics, and a way to track what she's covered. Price-sensitive; smartphone-first, sometimes low-bandwidth.

P2

Daniyal — final-year, exams & first job

Uses study material selectively; cares about the jobs portal and resume builder. Will apply for the Verified Pharmacist badge after licensure.

P3

Sana — hospital pharmacist, 7 yrs

Community-side persona. Reads and writes blogs (approved writer), answers Question threads, holds a Verified badge, browses senior roles occasionally.

P4

International pharmacy student

Discovers topics via search engines. Doesn't care about Pakistani semesters — validated by the subject-first taxonomy with semester as a subtle tag.

P5

Employer / HR at a pharma company

Submits job listings via a public form; interacts with the Jobs Manager, not the CMS.

P6

Internal content team

Authors, reviewers, blog editors, jobs manager, moderators, and admins working inside the CMS (§12).

4 Product Principles

1. **Content is the product.** Engineering exists to make excellent content easy to author, review, publish, find, and study.
2. **Free to read the preview, sign in for depth.** Public previews power SEO and discovery; the account unlocks full content and all study tools.
3. **Global by default, Pakistani by design.** Subject-first taxonomy, English UI, HEC semester numbers as metadata — with Pakistan-specific community and jobs value.
4. **Own the stack.** Self-hosted backend, MIT-licensed editor, components copied into the repo (shadcn model). No licensing landmines, no vendor lock-in.
5. **The user pays for their own AI.** BYOK keeps costs at zero, keeps the platform provider-neutral, and lets power users choose their model.
6. **Every feature ships with its admin surface.** Nothing goes to production without the CMS controls to manage it.
7. **Design for the phase you're in.** A 300-user community gets threads and categories, not vote-ranked feeds; automation replaces manual process only when volume demands it.

5 Decision Log

Eighteen product and technical decisions were locked through structured MCQ elicitation rounds before writing this document. This table is the contract; sections below implement it.

#	Decision	Choice	Notes / rationale
D1	Backend platform	**Supabase, self-hosted**	Delegated to PRD; full justification in §11.2.
D2	Rich-text editor	**Plate**	MIT end-to-end (incl. AI), shadcn-native distribution, 50+ plugins, comments/suggestions for editorial review. Runner-up BlockNote (XL/AI packages GPL-3.0 or paid). Yoopta: bus-factor risk; Novel: too thin.
D3	Business model	**Deferred — design for flexibility**	No paywall at launch. Entitlement + feature-flag layer from day one (§7.11, §16).
D4	Content hierarchy	**Subject → Chapter → Topic → Subtopic; HEC semester as subtle tag**	Keeps content usable worldwide; Pakistani students filter by semester when they want to.
D5	Progress tracking	**Manual "mark as complete" only**	At subtopic level; rolls up to topic/chapter/subject percentages.
D6	Self-assessment (MCQs)	**Later phase**	Scheduled Phase 5; can be pulled forward — Assumption A3.
D7	Auth methods	**Email + password, Google OAuth**	Via Supabase Auth (GoTrue); HTTPS prerequisite in §13.
D8	Content access	**Public previews; login for full content + features**	Preview = first section(s) of each topic, fully indexable.
D9	BYOK providers	**OpenAI · Anthropic · Gemini · OpenRouter · Groq**	Unified through a provider-adapter layer (§7.5).
D10	AI feature scope	**Explain + Summarize + Chat-with-topic**	Grounded strictly in ePharmatica content; ships in Phase 1.
D11	Pharmacist verification	**Optional badge, manual review**	Credential submission → admin review queue (§7.6.4).
D12	Forum model	**Hybrid**	Categories + threads backbone; Discussion/Question types; reply upvotes; accepted answers. Works at low volume.
D13	Blog publishing	**Approved writers only**	Writer application flow; editorial review before publish.
D14	Jobs sourcing	**Curated + employer submissions + aggregation later**	Run by a dedicated Jobs Manager CMS role; aggregation manual-first.
D15	WhatsApp system	**User-initiated content bot first**	Rides the free service-conversation window; paid outbound only after funding. Pricing re-verification note §7.9.
D16	Deployment	**Self-host on one VPS — Hetzner EU**	4 vCPU / 8 GB / 80–160 GB NVMe (CX33-class, Falkenstein/Nuremberg); Coolify + Docker; Cloudflare in front; offsite backups. Runbook §10.
D17	Phase 1 scope	**Learning + accounts + AI (BYOK) + landing**	
D18	Post-Phase-1 order	**Community → Jobs + Resume → WhatsApp bot**	AI ranked first but absorbed into Phase 1 (D17); remaining ranking defines Phases 2–4.

6 Phased Roadmap

Each phase has a hard gate: **acceptance criteria must pass before the next phase starts.** Timeboxes are planning estimates for a 1–2 developer team, not commitments.

P0 Foundation · 2–3 wks

Infra + skeleton. VPS provisioned with Coolify; self-hosted Supabase running with HTTPS; Cloudflare DNS/CDN; CI/CD green (lint, typecheck, test, build, deploy); Next.js app scaffolded with Tailwind v4 + shadcn/ui, design tokens, light/dark themes; Supabase migrations + type codegen wired; Sentry, Umami analytics, Uptime Kuma, nightly backups operational; **landing page live** with brand, feature overview, Coming Soon modules, and segmented waitlist capture (email + module + role); system tooling online — `/dev/system` primitives page, Storybook, Style Dictionary token pipeline, RTL-safe conventions from the first component.

ACCEPTANCE Deploy-on-merge works; backup restore drill passes on a scratch database; landing scores ≥ 90 Lighthouse performance/SEO/accessibility on mobile.

P1 Learning Platform MVP · 6–10 wks

Content model + CMS authoring (Plate) + review workflow; public learn pages with previews; accounts (email + Google); reader features: highlights, bookmarks, personal notes, mark-as-complete progress + dashboard; embedded short/long YouTube videos; site search (Postgres FTS); AI study suite (BYOK: explain / summarize / chat-with-topic); settings incl. AI key management.

ACCEPTANCE ≥ 2 subjects with ≥ 25 published subtopics end-to-end through the review workflow; a signed-in user can highlight, note, bookmark, complete, and use all three AI features with at least 2 providers; preview gating verified logged-out; RLS test suite passes.

P2 Community · 4–6 wks

Hybrid forum (categories, Discussion/Question threads, reply upvotes, accepted answers, reporting, moderation queue); blog module (approved writers, Plate editor, editorial workflow, public blog); Verified Pharmacist application + badge; user public profiles.

ACCEPTANCE Full moderation loop demonstrable (report → queue → action → audit log); a writer application → approval → draft → review → publish loop completes; badges render across forum/blog/profile.

P3 Jobs Portal + Resume Builder · 4–6 wks

Jobs listings (curated + employer submission form with review), filters/search, expiry automation, Jobs Manager CMS surface; resume builder: guided profile-driven data entry, ATS-safe default template + 2–3 styled templates, live preview, client-side PDF export, multiple saved resumes.

ACCEPTANCE An employer submission reaches publication only via Jobs Manager approval; expired jobs auto-hide; exported PDF passes a text-extraction check (selectable text, correct reading order) on the ATS template.

P4 WhatsApp Content Bot · 3–4 wks

Meta WhatsApp Business Cloud API webhook; menu-driven intents (Topics / Blogs / Jobs); topic search → summary + link (+ video link); latest blogs list; newest jobs list; session/message logging; admin visibility.

ACCEPTANCE End-to-end user-initiated conversation on a production number; zero business-initiated sends; logs visible in CMS.

P5 Assessment Engine · deferred · TBD

Practice MCQs per topic, attempt history, weak-area surfacing. Re-scope at Phase 3 planning (Assumption A3 — may be pulled earlier if learning-retention metrics demand it).

ACCEPTANCE Defined at re-scope.

CONTINUOUS TRACKS — ALL PHASES

Content production pipeline **and YouTube channel creation + publishing cadence** (the true critical path — see Risk R1; the channel does not yet exist and must be live with initial videos before Phase 1 launch), SEO, accessibility, performance budgets, security patching, and monthly restore drills.

7 Feature Specifications

7.1 Landing Page (Phase 0)

- Hero (S4): **split composition** — statement block (value proposition, brand mark, primary CTA **"Start learning free"** + ghost **"Join waitlist"**) beside a **live demo of real system primitives** (blister grid / mini reader); stacks statement-first on mobile.
- Feature sections for all six modules (Learning, AI Study Tools, Community, Blog, Jobs, Resume Builder); unreleased ones carry a **Coming Soon** ribbon and a waitlist form — **email + module of interest + role** (student / pharmacist / other) — stored in `waitlist_signups` (S12).
- Social proof placeholder, YouTube channel strip, FAQ, footer (About, Contact, Privacy, Terms, Content Policy).
- Fully static/ISR; Lighthouse ≥ 90 on mobile.

7.2 Learning Platform (Phase 1)

7.2.1 Content model

`Subject → Chapter → Topic → Subtopic`. The **subtopic is the atomic learning unit**: it owns the Plate JSON document, optional short-form video URL, optional long-form video URL, estimated reading minutes, and ordering. Topics aggregate subtopics; a topic with no natural subdivision gets a single default subtopic (UI hides the redundant level). Semester tagging: nullable `semester` (1–10) on topic (inheritable from subject default), displayed as a small tag/filter — never as primary navigation (D4). The HEC semester mapping is set and maintained by the content **Author** at authoring time (against the current HEC syllabus edition) and spot-checked during review (Q5 resolution, v1.1).

7.2.2 Authoring (CMS)

Plate editor configured with: headings, rich marks, lists, tables, callouts (Definition / Clinical Pearl / Mnemonic / Exam Alert / Warning), images (Supabase Storage, auto-compressed), YouTube embed block, code block, table-of-contents block, and a custom **Drug Card** block (name, class, MOA, uses, adverse effects, contraindications — structured fields rendered as a styled card). Slash menu + drag handles on. Comments/suggestions enabled for the review workflow.

7.2.3 Editorial workflow

`draft → in_review → changes_requested → approved → published` (+ `archived`). Authors write; Reviewers comment/suggest and approve; Publishers (Content Managers) publish. Every publish snapshots a revision (`content_revisions`) with diff-viewable history and one-click rollback. Audit log on all transitions.

7.2.4 Reader experience

Server-rendered topic pages: sticky sub-navigation of subtopics, right-rail table of contents (from headings), reading time, semester tag, prev/next navigation, breadcrumbs. Videos render as **lite YouTube facades** (thumbnail + play button; the iframe loads on click via YouTube's privacy-enhanced mode, `youtube-nocookie.com`) to protect performance and limit third-party cookies. Typography tuned for study: max ~70ch line length, generous spacing, print stylesheet.

7.2.5 Access gating (D8)

Logged-out users see the first N blocks (per-subtopic `preview_block_count`, default ≈ first section) followed by a fade-out and sign-in CTA. Full JSON for gated content is **never sent to the client** for anonymous sessions (server-side truncation, not CSS hiding). All interactive study tools require login.

7.2.6 Study tools (signed-in)

- **Highlights:** text-selection highlighting in 4 colors; stored as `{subtopic_id, block_id, anchor_offsets, text_snapshot, color}`; re-anchoring on content edits falls back to snapshot text search; orphaned highlights surface in “My Highlights” rather than silently vanishing.
- **Bookmarks:** any subtopic/topic; organized list in dashboard.
- **Personal notes:** a lightweight Plate instance attached per subtopic (private, RLS-enforced), plus notes attachable to a specific highlight.
- **Progress (D5):** explicit “Mark as complete” button per subtopic; completion timestamps; roll-up percentages per topic/chapter/subject; a subtle “Continue where you left off” card on the dashboard. No automatic tracking.
- **Dashboard:** continue-studying, progress by subject, recent highlights/notes, bookmarks.

7.2.7 Search

Postgres full-text search (`tsvector` over title + extracted plain text of published content), weighted (title > headings > body), with subject/semester filters. Upgrade path: Meilisearch/Typesense container when FTS relevance becomes limiting (not before).

7.3 Accounts & Profiles (Phase 1)

- Supabase Auth: email + password (with verification email) and Google OAuth (D7). Password reset, session management, account deletion (GDPR-style erasure incl. content anonymization rules).
- Profile: username (unique, URL-safe), display name, avatar, user type (`student | pharmacist | other`), institution (optional), country, bio. Public profile pages activate in Phase 2.
- Settings: profile, appearance (light/dark/system), AI keys (§7.5.4), data export (JSON of notes/highlights), delete account.

7.5 AI Study Suite — BYOK (Phase 1)

7.5.1 Providers (D9)

OpenAI, Anthropic, Google Gemini, OpenRouter, Groq — implemented via the Vercel AI SDK's provider abstraction behind a single internal `AiProvider` interface (streamText, model listing, error normalization). Users select provider + model; sensible defaults per provider. Provider docs are linked in-app (e.g., Anthropic's official API docs at docs.claude.com/en/api/overview) with short “how to get a key” guides — Groq and Gemini are surfaced first for students since they offer generous free tiers (verify current terms at build time).

7.5.2 Features (D10)

- **Explain:** select any text in a subtopic → “Explain with AI” → popover with explanation grounded in the surrounding section; follow-up “simpler / example / mnemonic” quick actions.
- **Summarize:** one-click structured summary of the current subtopic (key points, must-know terms, common exam angles).
- **Chat with topic:** side-panel chat whose context is the current topic's full content (serialized to Markdown). Strict grounding: the system prompt instructs answering from provided content, saying so when the answer isn't in it. Single-topic context fits comfortably in modern context windows; cross-topic RAG with `pgvector` is a future enhancement, not v1.
- Every AI surface carries a persistent disclaimer: educational aid, may contain errors, not medical advice — verify against authoritative sources.

7.5.3 Architecture

All calls go through a Next.js route handler that streams responses (SSE). The server injects the user's key per-request, never logs prompts/keys, and enforces per-user rate limits (e.g., 30 requests / 5 min) to protect users from key-burning bugs. No conversation storage in v1 (session-only), keeping the privacy posture simple; optional saved chats are a later enhancement.

7.5.4 Key management & security

Two storage modes, user-selectable: **“This device only” (default)** — key kept in the browser (IndexedDB), sent per-request over TLS to the proxy route, held in memory only; and **“Sync across devices”** — key encrypted server-side (libsodium sealed-box / AES-256-GCM) with an app-level encryption key held only in server env (never in the database), decrypted transiently per request. Keys are write-only in the UI (masked after save), validated with a 1-token test call on save, revocable instantly, and excluded from logs/Sentry via scrubbing rules. The data-deletion flow purges keys first.

7.6 Community Platform (Phase 2)

7.6.1 Forum — hybrid model (D12)

- **Structure:** fixed admin-managed categories (initial set: Study Help, Clinical Practice, Industry & Regulatory Affairs, Careers & Licensing, General Lounge, Platform Feedback), each holding threads.
- **Thread types:** `discussion` (open-ended) and `question`. Question threads display an answer count and allow the thread author (or a moderator) to mark one reply as the **Accepted Answer**, which pins to the top with a distinct style.
- **Votes:** upvotes on replies only (no downvotes — protects beginners). Default reply sort: accepted answer → most upvoted → chronological; user-switchable to pure chronological.
- **Composer:** a trimmed Plate configuration (rich text, lists, images, links, code) — the same editor family as everywhere else.
- **Signals & hygiene:** thread tags (free-form, admin-mergeable), pinned + locked threads, subscribed-thread in-app notifications, @mentions, edit windows with edit history, soft-delete.
- **Moderation:** report on any thread/reply (reason + note) → moderation queue in CMS → actions: dismiss, hide, delete, warn user, temp-mute (1/7/30 days), ban. All actions audit-logged. New-account friction: accounts under 24 h old are rate-limited (e.g., 2 threads + 10 replies/day) as basic spam control.
- **Verified Pharmacist badge (D11):** rendered beside the username in forum, blog, and profiles.

7.6.2 Blog (D13)

- Public blog with categories/tags, author pages, related posts, and full SEO treatment (OG images, structured data).
- **Approved writers only:** users apply via a form (motivation, sample, credentials) → application queue → approval grants the `blog_writer` role.
- Workflow mirrors study content: `draft → in_review → published` with Blog Editor review, revisions, scheduling (publish-at datetime), and cover images.
- Comments on posts are **off in v1** (community discussion belongs in the forum; a “Discuss in forum” button auto-creates/links a thread).

7.6.3 Public profiles (Phase 2)

`/u/[username]`: avatar, bio, badges, user type, joined date, recent forum activity, published blog posts. Study data (progress/notes/highlights) is never public.

7.6.4 Verified Pharmacist application

Form: full name, registration/license number, issuing council, graduation year, document upload (license/degree scan → private Storage bucket). Admin review queue with approve/reject + reason. Approved: badge + `verified_at`. Documents auto-purge 90 days after decision (privacy). No automated registry checks in v1 (Non-Goal).

7.7 Jobs Portal (Phase 3)

- **Listing schema:** title, organization, logo (optional), location (city + country + on-site/hybrid/remote), employment type (full-time / part-time / contract / internship / house job), category (hospital, community/retail, industry — production/QA/QC/RA/PV, sales & marketing, academia, regulatory bodies, other), experience level, salary range (optional, PKR-friendly), description (Plate), how-to-apply (external URL **or** email), source (`curated | employer_submission | aggregated`), posted/expiry dates, status (`pending | published | expired | rejected | archived`).
- **Sources (D14):** (a) Jobs Manager creates curated listings in CMS; (b) public employer submission form (with contact email verification) lands in a review queue; (c) later: aggregation from external boards/newspapers — manual-first via a quick-add CMS tool, automation deferred.
- **Public UX:** filterable index (category, city, type, experience, freshness), job detail pages with clear “Apply” out-links, share buttons (WhatsApp-first), and `JobPosting` schema.org structured data for Google Jobs indexing.
- **Lifecycle automation:** listings auto-expire (default 30 days, editable); expired listings 410/redirect to index with an “expired” notice.
- **Saved jobs** for signed-in users; **job alerts are out of v1** (arrive with the WhatsApp/notifications phase).

7.8 Resume Builder (Phase 3)

- **Data-driven:** users fill structured sections — personal/contact, professional summary, education, experience (incl. house job/internships), licenses & certifications, skills, publications/projects, references — stored as versioned JSON (`resumes` table, multiple resumes per user, duplicate-and-edit).
- **Templates:** `ATS Standard` (default: single column, semantic headings, standard fonts, zero graphics) + 2–3 styled templates (accent color, two-column). Template switch never touches data.
- **Export:** client-side PDF via `@react-pdf/renderer` — guarantees a true text layer (selectable, correct reading order) for ATS parsing; A4 default with US-Letter toggle.
- **Assists:** live preview, completeness meter, per-section writing tips (pharmacy-specific), optional AI polish of summary/bullets **using the user's BYOK key** (reuses §7.5 infrastructure; clearly labeled).
- **Privacy:** resumes are private; no public resume pages in v1; PDF generation happens in-browser (data never renders server-side).

7.9 WhatsApp Content Bot (Phase 4) — user-initiated (D15)

- **Model:** inbound-only assistant on the Meta WhatsApp Business Cloud API. Users message first; the bot replies within the customer-service window. Business-initiated template messaging (job alerts, content notifications, marketing) is explicitly **out of scope until funded** — the system stores opt-ins from day one so the upgrade is a switch-flip. *Note: Meta's WhatsApp pricing model has changed before (conversation-based → per-message); re-verify current free-tier mechanics for user-initiated service conversations at build time.*
- **Interaction design:** greeting → interactive list menu: **1) Find a topic 2) Latest blogs 3) New jobs 4) About ePharmatica.** Topic flow: user sends keywords → Postgres FTS lookup → top 3 matches as buttons → selected topic returns a short summary, the study link, and the short-form video link. Blogs/jobs flows return the latest 3–5 items with links. Unrecognized input → gentle fallback to menu. **No LLM in v1** (menus + FTS keep cost zero and behavior predictable); optional AI-assisted parsing later.
- **Architecture:** `/api/whatsapp/webhook` (signature-verified) → intent router → reply via Graph API. `wa_contacts` (phone hash, opt-in flags) + `wa_messages` log (90-day retention). CMS surface: conversation log viewer, canned-response editor, kill-switch.
- **Compliance:** respond only to inbound; honor STOP; publish a WhatsApp privacy note; phone numbers stored hashed except where reply routing requires the raw number (encrypted at rest).

7.10 CMS / Admin (built progressively, every phase)

A protected `/admin` area inside the same Next.js app (no separate deployment). Modules: Dashboard (content/publishing stats, queues) · Content (subjects/chapters/topics/subtopics tree, Plate editor, workflow queues, revisions) · Media library · Taxonomy (semesters, tags, forum categories, job categories) · Blog (posts, writer applications) · Community (moderation queue, reports, user sanctions) · Jobs (listings, submissions queue, expiry overrides) · Verifications (pharmacist badge queue) · Users (search, roles, sanctions) · WhatsApp (logs, canned responses) · Settings (feature flags, Coming-Soon toggles, entitlements, site metadata) · Audit log (filterable, exportable). Role-gated per §12; every mutating admin action writes to `audit_log`.

7.11 Monetization-Flexibility Layer (D3 — built in Phase 1, dormant)

- `plans` and `entitlements` tables + a single server-side `can(user, feature)` gate used by any feature that could ever be premium (AI surfaces, resume templates, future content tiers). At launch every user maps to an implicit `free` plan where everything current is allowed.
- Feature flags (DB-backed, cached) control module visibility incl. Coming Soon states — marketing toggles without deploys.
- Future payment note (not v1): Stripe is unavailable to Pakistan-based merchants; when monetization activates, evaluate merchant-of-record providers (Paddle, Lemon Squeezy) and local rails (JazzCash/Easypaisa via aggregators). Recorded here so architecture (server-checked entitlements, no client trust) stays gateway-agnostic.

8 Information Architecture & Routes



Slugs are stable, lowercase, hyphenated; renames create redirects (`slug_history`).

9 Data Model (Entity Overview)

All tables `snake_case`, UUID primary keys, `created_at`/`updated_at`, **row-level security enabled on every table** (§13.2). Plate documents are stored as `jsonb`. An ERD + full SQL migration set is a Phase 0 deliverable (Appendix C lists migration order).

Domain	Tables & key fields
Identity & access	`profiles` (1:1 auth.users — username, display_name, avatar_url, user_type, country, institution, bio, role, is_verified_pharmacist, verified_at) · `pharmacist_verifications` (license_no, council, document_path, status, reviewer_id, decided_at) · `plans` · `entitlements` · `feature_flags` · `waitlist_signups` (email, module, role) · `audit_log` (actor_id, action, entity, entity_id, meta jsonb)

Domain	Tables & key fields
Learning content	`subjects` (name, slug, description, icon, default_semester, order) · `chapters` (subject_id, name, slug, order) · `topics` (chapter_id, name, slug, semester, order, status) · `subtopics` (topic_id, title, slug, order, content jsonb, plain_text → tsvector, preview_block_count, short_video_url, long_video_url, reading_minutes, status, published_at) · `content_revisions` (subtopic_id, content jsonb, editor_id, note) · `content_review_events` (from_status, to_status, actor_id, comment)
Study data (owned by user_id)	`highlights` (subtopic_id, block_id, anchor jsonb, text_snapshot, color, orphaned) · `bookmarks` (target_type, target_id) · `user_notes` (subtopic_id, highlight_id?, content jsonb) · `progress` (subtopic_id, completed_at; unique(user_id, subtopic_id))
AI	`ai_keys` (provider, encrypted_key — null in device-only mode, storage_mode, last_validated_at) · `ai_rate_limits` (rolling counters)
Community	`forum_categories` · `threads` (category_id, author_id, type, title, slug, content jsonb, accepted_reply_id, is_pinned, is_locked, reply_count, last_activity_at) · `replies` (thread_id, author_id, content jsonb, upvote_count, is_hidden) · `reply_votes` (unique(user_id, reply_id)) · `thread_tags` / `thread_tag_links` · `thread_subscriptions` · `reports` (target_type, target_id, reporter_id, reason, status, resolver_id) · `user_sanctions` (type, expires_at, reason)
Blog	`blog_posts` (author_id, title, slug, excerpt, cover_path, content jsonb, category_id, status, scheduled_at, published_at) · `blog_categories` · `writer_applications` (motivation, sample_url, status)
Jobs & resume	`jobs` (per §7.7 incl. source, status, expires_at) · `job_submissions` (raw employer form + verification token) · `saved_jobs` · `resumes` (title, data jsonb, template_id — templates defined in code)
WhatsApp	`wa_contacts` (phone_hash, phone_encrypted, opt_in_alerts) · `wa_messages` (contact_id, direction, payload jsonb, intent)

Key relationships: subjects 1—N chapters 1—N topics 1—N subtopics; subtopics 1—N revisions/highlights/notes/progress; threads 1—N replies; users 1—N everything they own.

10 System Architecture & Infrastructure (D16)

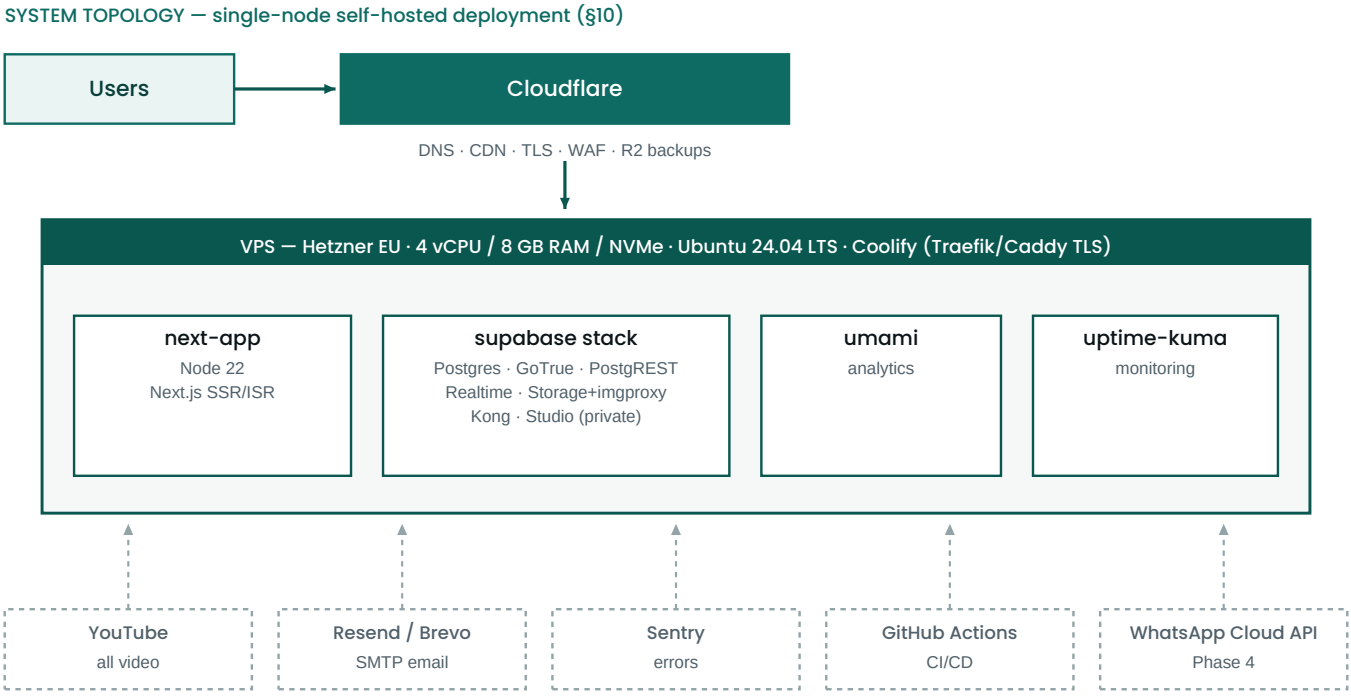


Figure 2 — Single-VPS topology. Studio and the Supabase dashboard bind to a private admin hostname, IP-allow-listed.

10.1 Sizing rationale (researched July 2026)

The self-hosted Supabase stack runs ~12 containers and is memory-bound; community/production guidance converges on **8 GB RAM + 4 cores + NVMe**, with 4 GB viable only when optional services are disabled. Reference cost: Hetzner CX33 (4 vCPU / 8 GB) ≈ €6.49/mo post-April-2026 pricing, 20 TB traffic included in EU regions; comparable specs elsewhere run ~$12–25/mo. Region (decided, v1.1): **Hetzner EU — Falkenstein or Nuremberg** — best price-per-spec; the ~120–150 ms path to Pakistan is mitigated by Cloudflare edge caching of public pages. Revisit only if interactive-latency metrics later demand it. Video bandwidth is entirely on YouTube.

10.2 Ops runbook (musts)

- **Backups:** nightly `pg_dump` (custom format) + weekly base backup + Storage bucket sync → encrypted upload to Cloudflare R2 / Backblaze B2; 30-day retention; **monthly restore drill** into a scratch container (calendar-enforced). Targets: RPO ≤ 24 h, RTO ≤ 4 h.
- **Security baseline:** SSH keys only + fail2ban; UFW default-deny (80/443 + SSH); unattended-upgrades; Docker images pinned + monthly patch window; secrets only in the Coolify env store; HTTPS everywhere (hard requirement for Google OAuth); Supabase JWT secret + keys rotated on any suspicion.
- **Upgrades:** self-hosted Supabase upgrades are manual — test on a scratch clone first, snapshot before, maintenance window announced in-app.
- **Scale path:** vertical resize → split Postgres to its own VPS → add a second app node behind a Cloudflare load balancer. No Kubernetes.

11 Technology Stack — Decisions & Justification

11.1 Stack summary

Layer	Choice
Framework	**Next.js 16** (latest 16.2.x — App Router, RSC, Turbopack default), TypeScript 5 strict, Node 22 LTS, pnpm
UI	**Tailwind CSS v4 + shadcn/ui** (components owned in-repo), lucide icons, dark/light via next-themes
Editor	**Plate** (installed via shadcn registry) — one editor family across study content, blog, forum, notes
Backend	**Supabase (self-hosted)**: Postgres, GoTrue Auth, PostgREST, Storage (+imgproxy), Realtime (used sparingly)
Data layer	supabase-js + TanStack Query v5; Server Actions / route handlers for privileged mutations
Forms / validation	react-hook-form + Zod (schemas shared client/server)
AI	Vercel AI SDK provider adapters (OpenAI · Anthropic · Gemini · OpenRouter · Groq), SSE streaming proxy
PDF	@react-pdf/renderer (client-side)
Search	Postgres FTS (tsvector + websearch_to_tsquery)
Quality	ESLint (flat) + **Prettier** + TS strict + Husky/lint-staged + Conventional Commits
Testing	Vitest + Testing Library (unit) · Playwright (e2e smoke) · SQL-based **RLS policy tests**
Codegen	`supabase gen types typescript` on every migration (CI drift check); Zod schema inference
Observability	Sentry (client + server, PII scrubbed) · Umami · Uptime Kuma · structured pino logs
CI/CD & hosting	GitHub Actions → Coolify webhook deploy on the VPS (§10, §13.4)

11.2 Why Supabase over Nhost (D1)

Both are open-source BaaS with Postgres, auth, storage, and functions, and both self-host via Docker. Supabase wins for ePharmatica on five grounds:

- **Ecosystem & longevity** — a far larger community, documentation base, and third-party tooling, which matters when one small team owns ops for years and hires later.
- **Access-model fit** — the app is a straightforward CRUD + RLS system; PostgREST + supabase-js serves it directly, whereas Nhost's core value (Hasura GraphQL) adds a GraphQL layer, permission DSL, and an extra always-on service we don't need — meaningful RAM on an 8 GB box.
- **Codegen requirement** — `supabase gen types typescript` yields end-to-end typed queries with zero schema drift, integrating natively with TanStack Query; the GraphQL-codegen equivalent is possible but heavier.
- **Postgres-native security** — RLS policies live in migrations as reviewable SQL and are testable in CI (§13.2), rather than in Hasura metadata.
- **Team gravity** — Supabase knowledge is dramatically easier to hire/contract for in 2026.

Nhost remains a respectable alternative if the product ever becomes GraphQL-first; nothing in the schema precludes migration since both are plain Postgres underneath.

11.3 Why Plate (D2 — recap)

MIT across everything used (including AI integration — no copyleft or paid tiers on the critical path, unlike BlockNote XL); components installed via the shadcn CLI so the editor is owned, restyled, and versioned inside the repo exactly like the rest of the design system; the largest plugin surface for the custom blocks this product needs (Drug Cards, callout taxonomy, embeds); built-in comments/suggestions powering the editorial review workflow; and a single serialization (Plate JSON) for study content, blogs, forum posts, and private notes. Accepted trade-off: a steeper learning curve (Slate concepts, serialization discipline) — mitigated by pinning versions, snapshot tests on renderers, and a thin internal `@epharmatica/editor` wrapper so product code never touches Slate directly.

12 Roles & Permissions

Application roles live in `profiles.role` (single primary role) enforced by RLS + a server-side policy map; one account may additionally hold the `blog_writer` capability and the Verified Pharmacist badge (orthogonal flags). Rules: least-privilege defaults; no role can edit published study content without a new revision; all admin mutations are audit-logged; role changes require `admin`+ and are themselves audited.

Role	Capabilities (cumulative summary)
`user (default)`	All learner features; forum participation; blog reading; job saving; resume builder; verification & writer applications
`blog_writer (capability)`	+ create/edit own blog drafts, submit for review
`author`	+ create/edit study content drafts, submit for review, view own revision history
`reviewer`	+ review queues: comment/suggest, approve or request changes on study content
`content_manager`	+ publish/unpublish/archive study content, manage taxonomy & media, rollback revisions
`blog_editor`	+ review/publish blog posts, manage blog categories, decide writer applications
`moderator`	+ moderation queue, hide/delete community content, warn/mute/ban users, manage forum categories/tags
`jobs_manager`	+ full jobs CRUD, approve/reject employer submissions, expiry overrides
`support`	+ read-only user lookup, resolve verification queue (approve/reject badges)
`admin`	+ user role assignment (below admin), feature flags & settings, WhatsApp console, full audit log
`super_admin`	+ everything, incl. admin management and destructive operations (confirmation + audit protected)

13 Non-Functional Requirements & Engineering Standards

13.1 Performance

Budgets on mid-range Android over throttled 4G (the realistic Pakistani baseline): topic pages **LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1**; JS ≤ 200 KB gz on reader routes (editor/admin bundles code-split away from public routes). Techniques: ISR for all public content (revalidate on publish via tag invalidation), Cloudflare edge caching, `next/image` with sharp, lite-YouTube facades, font subsetting, streaming RSC. Lighthouse CI check on landing + one reader page in the pipeline (warn < 90).

13.2 Security

- **RLS on every table, deny-by-default**, with policy conventions: owner-only for study data/keys/resumes; published-only public reads for content; role-checked admin access via a `has_role()` SQL helper. **CI runs an RLS test suite** (SQL tests against a migrated shadow DB asserting e.g. “anon cannot read unpublished subtopics”, “user A cannot read user B's notes/keys”).
- Service-role key **never** leaves server context; privileged mutations only via route handlers / server actions with Zod-validated input.
- Standard headers (CSP allowing YouTube embeds, HSTS, frame-ancestors), CSRF-safe patterns, upload constraints (type/size sniffing; private buckets for verification docs), per-route rate limiting (auth, AI proxy, submissions, webhook) + Cloudflare rules.
- Secrets in Coolify env only; Sentry PII scrubbing (keys, phone numbers, emails); dependency audit (`pnpm audit` + Renovate) weekly.
- AI-specific: prompt-injection posture — model output is rendered as text/markdown only, never executed or granted tool access; user keys handled per §7.5.4.

13.3 SEO, accessibility, i18n

Semantic HTML + schema.org (`Article`, `JobPosting`, `BreadcrumbList`), per-route metadata + OG image generation, sitemaps (content, blog, jobs) + robots, canonical URLs, `slug_history` redirects. **WCAG 2.1 AA:** full keyboard support (incl. editor fallbacks), visible focus, contrast-checked tokens in both themes, reduced-motion respect, alt-text required on CMS image upload. **i18n:** English-only UI at launch but all strings routed through next-intl from day one; Urdu (with an RTL-aware layout audit) is the designated second locale (Assumption A2).

13.4 Environments, CI/CD, codegen

- **Environments:** `local` (Supabase CLI stack + seed script with demo content/users per role) → `staging` (separate Supabase project/compose on the same VPS, basic-auth-gated, deploys from `develop`) → `production` (deploys from `main`).
- **Pipeline (GitHub Actions):** PR → install (pnpm cache) → Prettier check → ESLint → `tsc --noEmit` → **type-drift check** (regenerate Supabase types; fail on diff) → Vitest → build → Playwright smoke (auth, reader, gating). Merge to `main` → build + push image → Coolify deploy webhook → post-deploy: gated migration step with automatic pre-migration dump → smoke ping → Sentry release + sourcemaps.
- **Migrations:** Supabase CLI migration files in-repo; forward-only; every migration paired with regenerated types and, where relevant, RLS tests. **Branching:** trunk-based (`main` protected, short-lived feature branches, Conventional Commits, squash merge).

14 Success Metrics (first 6 months post-Phase-1)

NORTH-STAR METRIC

Weekly Active Learners — users completing ≥ 1 subtopic or creating ≥ 1 highlight/note in a week.

- **Acquisition:** 5,000 registered accounts; ≥ 40% of new sign-ups from organic search / YouTube referral.
- **Activation:** ≥ 50% of new accounts mark ≥ 3 subtopics complete in week 1.
- **Retention:** ≥ 25% week-4 retention among activated learners.
- **Content:** ≥ 150 published subtopics across ≥ 6 subjects; median review cycle < 7 days.
- **AI adoption:** ≥ 20% of WAL configure a BYOK key; explain / summarize / chat usage tracked per feature (Umami events).
- **Later phases add:** threads/week & accepted-answer rate (P2) · published posts/month (P2) · live listings & apply-outs (P3) · resumes exported (P3) · WhatsApp sessions/week (P4).

All measured via self-hosted Umami events + SQL dashboards; no third-party trackers. Targets are directional, reviewed monthly.

15 Risks & Mitigations

#	Risk	Impact	Mitigation
R1	Content production bottleneck — engineering outpaces authored, reviewed syllabus content	*Empty product*	Treat content as a parallel workstream with its own pipeline metrics (§14); seed 2 complete subjects before public launch; recruit student authors under reviewer supervision; Phase 1 acceptance gates on content, not just code.
R2	Single-VPS / single-operator failure (outage, upgrade gone wrong, bus factor)	*Downtime, data loss*	§10 runbook: tested backups + monthly restore drills, snapshots before upgrades, Uptime Kuma alerts, documented recovery playbook; modest HA gap accepted consciously at this stage.
R3	BYOK friction — students won't fetch API keys	*AI features unused*	First-run wizard with screenshots per provider; order Groq/Gemini (free tiers) first; the product remains fully valuable without AI.
R4	Medical-education accuracy errors (authored or AI)	*Trust / harm*	Mandatory reviewer approval before publish (G6); source-citation norms in the content style guide; persistent AI disclaimers + strict grounding (§7.5.2); error-report button on every subtopic.
R5	Moderation load / spam as the community opens	*Quality decay*	New-account rate limits, report→queue→sanction loop from day one of Phase 2, category-scoped moderators, content policy published before launch.
R6	Plate editor complexity (serialization bugs, upgrades)	*Content corruption*	Version-pinned editor wrapper package, JSON schema validation on save, renderer snapshot tests, revisions provide rollback.
R7	Platform dependencies shift (YouTube embed policies, Meta WhatsApp pricing)	*Feature / cost break*	Video URLs stored abstractly (player swap possible); WhatsApp phase gated on pricing re-verification (§7.9); no business logic depends on either being free forever.
R8	Monetization deferred too long / retrofit pain	*Runway*	Entitlement gate (§7.11) wired through all premium-candidate surfaces from Phase 1; quarterly business-model review checkpoint.

16 Assumptions & Open Questions

16.1 Assumptions (proceed unless overridden)

- **A1:** Web-only responsive app; PWA-ready but no install campaign in v1.
- **A2:** English UI at launch; next-intl wiring from day one; Urdu is locale #2 (timing open).

- **A3:** Assessment engine sits at Phase 5; revisit at Phase 3 planning (D6).
- **A4:** Resume PDF is client-side `@react-pdf/renderer`; the ATS template is the default.
- **A5:** AI chats are session-only in v1 (no server-side conversation storage).
- **A6:** Blog comments off; discussion routed to forum threads.
- **A7 — RESOLVED (v1.3–v1.5):** Design system locked in STYLE.md v1.3 — **Pharmacopoeia** palette (teal #0E6B63 on archival #FBFBF9), **0px radius (final)**, Archivo / Instrument Sans / IBM Plex Mono, monograph-label signature, mobile & touch system, decisions S1–S12. This PDF adopts the locked palette; its typeface remains the interim print pairing.
- **A8:** Initial launch subjects: Pharmacology + Pharmaceutics (highest-demand, video-friendly) — confirm with content team.

16.2 Open questions (status as of v1.6)

#	Status	Detail
Q1	RESOLVED	Domain is **epharmatica.com**. The YouTube channel does not yet exist; channel creation, branding, and an initial video pipeline are a Phase 0–1 content-track deliverable (§6). Release cadence commitment still to be defined by the content team.
Q2	OPEN	Content team shape at launch (authors/reviewers count) — drives Phase 1 duration more than engineering does.
Q3	RESOLVED	**Hetzner EU** (Falkenstein/Nuremberg, CX33-class). See §10.
Q4	RESOLVED (drafts)	Legal pages drafted as companion documents — Terms of Service, Privacy Policy, Content Policy, Medical Disclaimer. Entity name, registered address, and governing-city fields remain placeholders; **review by qualified counsel is required before public launch.**
Q5	RESOLVED	Semester-tag mapping is owned by the authoring **Author** (against the current HEC Pharm-D syllabus edition), verified at review (§7.2.1).

17 Appendices

A. Environment variables (initial set)



B. Glossary

Subtopic — atomic learning unit owning one Plate document. **Preview** — the public, indexable head of a subtopic. **BYOK** — bring-your-own-key; user-supplied AI provider credentials. **Accepted Answer** — reply marked as resolving a Question thread. **Entitlement** — server-checked capability grant, monetization-ready. **Verified Pharmacist** — manually reviewed credential badge.

C. Migration order (Phase 0/1 outline)

001 extensions (pgcrypto, pg_trgm) + helpers (`has_role()`) → 002 profiles + auth trigger → 003 plans/entitlements/feature_flags → 004 content tree (subjects…subtopics) → 005 revisions + review events + slug_history → 006 study data (highlights, bookmarks, notes, progress) → 007 ai_keys + rate limits → 008 waitlist + audit_log → 009 FTS columns/indexes/triggers → 010 storage buckets + policies. Community / jobs / resume /

WhatsApp migrations land in their phases. Every migration ships with RLS policies + tests + regenerated types (§13.4).

End of PRD v1.6 — living document; amend via PRs with changelog entries. · ePharmatica · epharmatica.com